

TARKO-SALE, Russian Federation

WMO: 235520

Lat: 64.9114N

Lon: 77.7708E

Elev: 89

StdP: 14.65

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.8	-42.0	-50.2	0.3	-45.3	-45.9	0.4	-41.3	17.1	6.7	14.9	8.8	2.8	70	0.726

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	82.7	66.4	78.4	64.1	74.3	62.2	67.9	78.6	65.7	75.5	63.5	72.3	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	89.9	72.4	61.9	82.9	70.0	59.5	76.2	68.5	32.3	78.7	30.6	75.6	28.9	72.4	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	16.2	13.8	12.0	DB	-48.8	86.8	7.6	3.4	-54.3	89.2	-58.7	91.2	-63.0	93.1	-68.6	95.6
(5)				WB	-47.3	71.1	6.2	3.3	-51.7	73.5	-55.3	75.4	-58.8	77.2	-63.3	79.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.2	-12.5	-8.0	8.3	19.3	33.8	54.5	61.4	54.5	42.6	25.9	2.0	-6.1
	DBStd	28.67	17.66	16.73	14.62	13.39	10.99	10.98	8.21	7.05	8.06	11.46	17.41	18.19
	HDD50	10582	1938	1625	1293	920	517	80	5	29	249	746	1440	1740
	HDD65	15376	2403	2045	1758	1370	968	347	172	336	671	1211	1890	2205
	CDD50	782	0	0	0	0	15	213	358	168	28	0	0	0
	CDD65	101	0	0	0	0	1	30	60	10	0	0	0	0
	CDH74	932	0	0	0	0	6	299	541	85	1	0	0	0
	CDH80	230	0	0	0	0	2	65	151	12	0	0	0	0
Wind	WSAvg	5.6	5.0	5.2	5.9	6.4	6.7	6.1	5.6	5.1	5.4	5.8	5.3	5.2
Precipitation	PrecAvg	20.2	1.1	0.9	1.1	1.2	1.5	2.0	2.4	3.0	2.3	2.1	1.5	1.2
	PrecMax	30.0	2.0	2.0	2.4	2.5	3.4	4.5	6.5	5.9	4.6	4.9	2.8	2.4
	PrecMin	14.6	0.3	0.1	0.2	0.2	0.6	0.2	0.1	0.5	0.7	0.8	0.5	0.6
	PrecStd	3.2	0.4	0.4	0.5	0.6	0.7	1.0	1.3	1.4	1.0	0.7	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.3	29.2	36.9	49.8	72.0	86.1	88.5	81.4	69.6	52.0	36.0	31.5
		MCWB	26.4	28.2	32.9	41.7	57.4	66.2	68.0	69.3	59.2	47.8	34.8	30.6
	2%	DB	22.0	24.5	33.3	44.1	63.1	81.7	84.3	75.9	63.6	46.3	33.5	27.2
		MCWB	21.2	23.3	30.8	38.1	52.7	65.3	66.5	64.6	56.2	44.1	32.7	26.3
	5%	DB	16.2	20.3	31.0	40.5	56.4	77.6	80.0	70.8	59.0	42.8	30.8	24.0
		MCWB	15.5	19.3	29.0	35.8	48.3	63.1	64.8	61.7	53.9	40.8	29.9	23.3
	10%	DB	11.8	15.8	28.1	37.7	50.4	72.8	75.9	66.4	54.9	39.7	27.0	19.5
		MCWB	11.2	14.9	26.3	34.4	44.5	60.8	63.2	59.7	50.7	38.1	26.1	18.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	28.6	33.5	42.1	58.0	69.3	71.2	70.7	61.6	48.6	35.1	30.7
		MCDB	27.0	29.2	35.7	47.8	68.2	81.2	83.1	79.4	68.2	51.4	35.7	31.4
	2%	WB	21.2	23.6	31.3	38.5	53.6	66.9	68.3	65.9	57.2	44.3	32.9	26.5
		MCDB	22.0	24.7	33.1	43.0	61.9	78.0	80.0	72.6	62.3	46.0	33.6	27.1
	5%	WB	15.7	19.5	29.1	36.5	49.1	64.5	66.3	62.9	54.3	41.1	29.9	23.2
		MCDB	16.2	20.4	30.8	39.9	56.0	75.6	76.8	69.6	58.9	42.8	30.7	24.0
	10%	WB	11.2	14.9	26.3	34.6	44.5	61.9	64.4	60.3	50.9	38.1	26.1	18.8
		MCDB	11.9	15.7	28.2	37.4	50.2	71.2	73.8	65.9	54.5	39.7	27.1	19.5
Mean Daily Temperature Range	5% DB	MDBR	13.2	14.3	16.9	17.3	13.9	15.7	15.8	13.7	11.1	9.2	12.5	13.3
		MCDBR	16.3	15.1	16.0	14.1	22.3	22.0	21.9	20.8	18.0	10.2	13.0	13.0
		MCWBR	16.2	14.6	14.2	10.1	13.6	9.6	9.0	10.8	10.8	8.2	13.0	13.0
	5% WB	MCDBR	16.4	15.2	15.7	14.0	21.7	21.2	19.3	18.7	16.9	9.8	12.9	13.3
		MCWBR	16.4	14.9	14.5	10.8	14.0	10.8	8.9	10.5	10.8	8.2	13.0	13.3
Clear-Sky Solar Irradiance	taub	0.213	0.272	0.324	0.342	0.358	0.375	0.418	0.397	0.338	0.292	0.208	0.219	
	taud	1.876	2.026	2.001	2.058	2.221	2.398	2.260	2.290	2.418	2.196	1.905	1.881	
	Ebn at Noon	86	188	229	258	267	266	247	240	234	182	85	19	
	Edh at Noon	8	20	33	40	38	33	36	32	23	17	7	3	
All-Sky Solar Radiation	RadAvg	45	267	788	1453	1727	1721	1569	1063	619	219	72	15	
	RadStd	3	21	50	138	132	128	179	129	90	24	11	1	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+2.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+2.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page